Universal gravitation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Predict how gravitational force changes with mass and separation.

SUCCESS CRITERIA - I can...

1. Use center-to-center separation.
2. Apply direct mass and inverse-square distance comparisons.
3. Identify equal forces on the two interacting bodies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “separation” to explain your model or result. Sentence starters are optional.

gravity: an attractive interaction between masses

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A rover travels around a bend at a steady 5 m/s. **Describe** an observation that shows velocity changes even though the speed reading does not.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

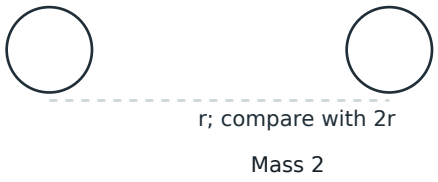
02 Worked example: explain the model

Reference: $F = Gm_1m_2/r^2$.
r is center-to-center separation. At fixed masses, doubling r makes the attractive force one quarter as large.

Worked reasoning: If separation doubles while both masses stay fixed, gravitational force becomes $1/2^2=1/4$ as large. A 20 N interaction would become 5 N.

Your turn: Sketch the two separations using centers. **Draw** equal and opposite force arrows for each pair.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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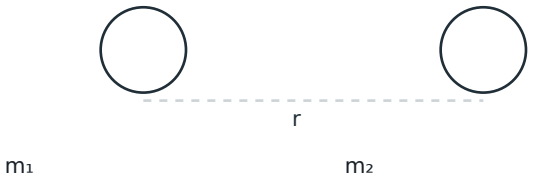
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two equal spheres are shown.
Mark the distance r used by the gravity equation and draw a force on each sphere.

Sentence starter: The … in my model represents … . Its label shows … .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

A gravity simulator shows a 12 N attraction. Run two separate comparisons from that starting setup: (a) double one mass; (b) double the center-to-center distance. Keep everything else fixed in each trial. Complete the force column.

Sentence starter: I notice … . This supports … because … .

Trial	Change from start	Force (N)
Start	None	12
A	One mass ×2	____
B	Distance ×2	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

Does a large planet exert more force on a small moon than the moon exerts on the planet?
Compare forces and accelerations.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A runner pushes the ground backward with 60 N.
Name the partner force, its size and the object it acts on.
Explain its role in the runner's motion.
Put each partner force on the object it acts on.

Sentence starter: My claim is … . The diagram/data show …, so … .



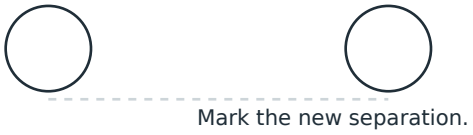
Object A Object B

Marzano check - circle: 1 2 3

07 Exit check

At separation r the force is 8 N. At 2 r with masses unchanged, what is it?

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Starting from a force of 12 N, double one mass and double the separation.
Find the new force and explain both effects separately.

Sentence starter: My claim is … . The diagram/data show …, so … .

Quantity	Original	New
m_1	m_1	____
m_2	m_2	____
r	r	____
Force	F_0	____

Marzano check - circle: 1 2 3